BLOCK-003 Derivation v5:

Normalization Principle Choices to Fix C in $\kappa_5^2 = C \cdot \sigma^{-3/4}$

EDC Working Document

2026-02-02

Abstract

Derivation v4 proved a No-Go Lemma: the EDC parameters $\{\sigma, \rho_P, R_\xi\}$ provide only two independent scales, insufficient to fix the dimensionless constant C in $\kappa_5^2 = C\sigma^{-3/4}$. This note does not close BLOCK-003; instead, it isolates the missing normalization choice by documenting exactly two candidate principles: **NP1** (EDC-internal postulate [P]) fixes C by identifying $\ell_\sigma = \sigma^{-1/4}$ as the fundamental 5D gravitational length; **NP2** (calibrated closure [Cal]) fixes C by a single calibration to the observed Newton constant G_N^{obs}. Both options are presented with their consequences for G_N in the compact extra dimension scenario.

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1 Restatement of the P15 No-Go Result

Lemma 1 (Dimensional Counting No-Go, from v4)**.** *Let the available EDC inputs be* $\{\sigma, \rho_P, R_\xi\}$ *with the pressure-balance constraint* $\sigma \sim \rho_P \cdot R_\xi^n$*. Then:*

1. *The constraint reduces three parameters to two independent scales.*
2. *The dimensionless* $C = \kappa_5^2 \sigma^{3/4}$ *requires three independent scales to be uniquely determined.*
3. *Therefore,* C *cannot be fixed from* $\{\sigma, \rho_P, R_\xi\}$ *alone without an additional normalization principle.*

Status: Proven in derivation v4. This note assumes the No-Go and explores resolution paths.

2 The Normalization Fork

The No-Go lemma implies that closing κ_5^2 requires exactly one additional input. We identify two clean options:

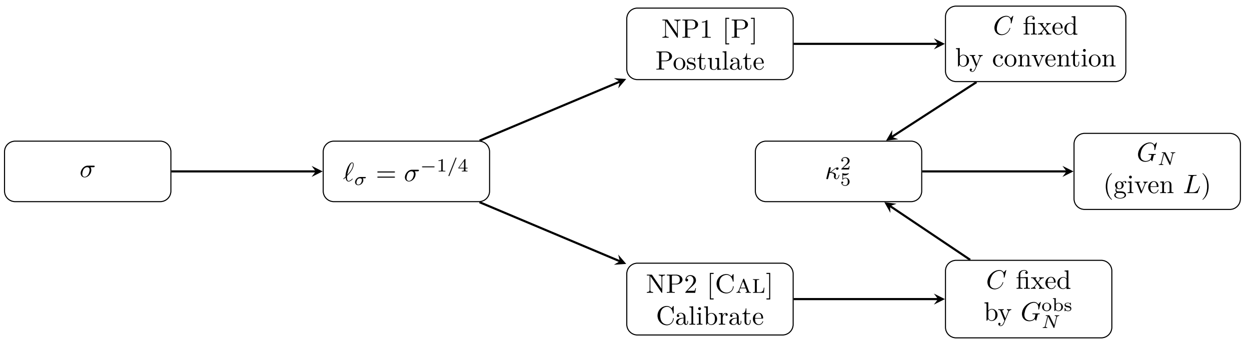

Figure 1: Schematic of the normalization fork. From brane tension σ, the length scale ℓ_σ emerges. Two principles can fix the dimensionless C: NP1 by postulate, NP2 by calibration. Either path yields κ_5^2, which combined with compactification length L gives G_N. *Schematic; not a result.*

3 NP1: EDC-Internal Postulate

Normalization Principle 1 (NP1: Identification Postulate [P])**.** The tension-derived length scale $\ell_\sigma = \sigma^{-1/4}$ is identified as the fundamental 5D gravitational length ℓ_5, with the standard gravitational normalization:

$$\kappa_5^2 = 8\pi\ell_5^3 = 8\pi\ell_\sigma^3 = 8\pi\sigma^{-3/4} \tag{1}$$

3.1 Consequences of NP1

Under NP1, the constant C is fixed by convention:

$$\boxed{C = 8\pi} \tag{2}$$

The 4D Newton constant (for compact extra dimension of size L) becomes:

$$G_N = \frac{\kappa_5^2}{6\pi L} = \frac{8\pi\sigma^{-3/4}}{6\pi L} = \frac{4}{3} \cdot \frac{\sigma^{-3/4}}{L} \tag{3}$$

3.2 Epistemic Status

- **Tag:** [P] (postulate, not derived)
- **What it assumes:** ℓ_σ is the natural 5D Planck length in EDC; the factor 8π follows standard GR normalization convention.
- **What remains open:** The compactification length L must still be specified. If $L = R_\xi$ [P], then:

$$G_N^{\mathrm{NP1}} = \frac{4}{3} \cdot \frac{\sigma^{-3/4}}{R_\xi} \quad \text{[P]} \tag{4}$$

3.3 Predictivity

Under NP1 + $L = R_\xi$, G_N is determined by σ and R_ξ alone. The model becomes predictive *if* σ can be computed from EDC field equations (currently [Open]).

4 NP2: Calibrated Closure

Normalization Principle 2 (NP2: Single Calibration [Cal])**.** The constant C is fixed by requiring that the resulting G_N matches the observed value $G_N^{\mathrm{obs}} = 6.674 \times 10^{-11}\,\mathrm{m^3 kg^{-1} s^{-2}}$ for a specified L.

4.1 Consequences of NP2

From $G_N = \kappa_5^2/(6\pi L)$ and $\kappa_5^2 = C\sigma^{-3/4}$:

$$G_N^{\mathrm{obs}} = \frac{C\sigma^{-3/4}}{6\pi L} \tag{5}$$

Solving for C:

$$\boxed{C = 6\pi L \cdot G_N^{\mathrm{obs}} \cdot \sigma^{3/4}} \tag{6}$$

4.2 Epistemic Status

- **Tag:** [Cal] (calibrated, not first-principles)
- **What it assumes:** One empirical input (G_N^{obs}) is used to fix the normalization. This is operationally legitimate but not a derivation.
- **What remains open:** The compactification length L appears in the calibration. If $L = R_\xi$ [P], then:

$$C = 6\pi R_\xi \cdot G_N^{\mathrm{obs}} \cdot \sigma^{3/4} \quad \text{[Cal]} \tag{7}$$

4.3 Predictivity

Under NP2, G_N is *not* predicted—it is input. However, the calibration makes other gravitational observables (e.g., corrections to Newtonian potential, brane-world signatures) predictive, conditional on the model structure.

Key distinction: NP2 does not claim to derive G_N; it uses G_N^{obs} once to close the system, then explores consequences.

Aspect	**NP1 (Postulate)**	**NP2 (Calibration)**
Tag	[P]	[Cal]
Fixes C by	Convention (8π)	Empirical input (G_N^{obs})
G_N status	Predicted (if σ known)	Input
Falsifiable?	Yes (if σ computed)	No (by construction)
EDC-internal?	Yes	Partially (uses external data)
Requires L?	Yes (e.g., $L = R_\xi$ [P])	Yes (enters calibration)

Table 1: Comparison of normalization principle options.

5 Comparison Table

6 What This Note Does Not Do

This note explicitly does **not**:

1. **Close BLOCK-003.** The gravity sector remains open because neither NP1 nor NP2 is derived from EDC field equations.
2. **Derive C from first principles.** Both options involve either postulating a convention (NP1) or using empirical input (NP2).
3. **Specify L.** The compactification length remains symbolic unless separately postulated as $L = R_\xi$ or derived from EDC geometry.
4. **Prefer one option over the other.** The choice between NP1 and NP2 is a modeling decision, not a mathematical necessity.

What this note does: It isolates the exact missing element (one normalization principle) and presents the two cleanest options available within the current EDC framework.

7 Path Forward

To truly close BLOCK-003 without postulate or calibration, one would need:

1. **A derived normalization** [Dc]: An EDC mechanism that produces a specific numerical prefactor (e.g., topological quantization, $\mathbb{Z}_6$ symmetry constraint, or holographic bound).
2. **A derived L** [Dc]: An EDC equation that determines the compactification scale from internal geometry (e.g., L from membrane thickness δ or bulk curvature).

Until such derivations exist, NP1 or NP2 represent the available options for operational closure.

8 Summary

Outcome: Two normalization principle options documented.

- **NP1** [P]: $\kappa_5^2 = 8\pi\sigma^{-3/4}$ (convention) $\Rightarrow G_N = \frac{4}{3}\sigma^{-3/4}/L$
- **NP2** [Cal]: $C = 6\pi L \cdot G_N^{\mathrm{obs}} \cdot \sigma^{3/4}$ (calibration) $\Rightarrow G_N = G_N^{\mathrm{obs}}$ by construction

BLOCK-003 remains OPEN pending a derived normalization principle.